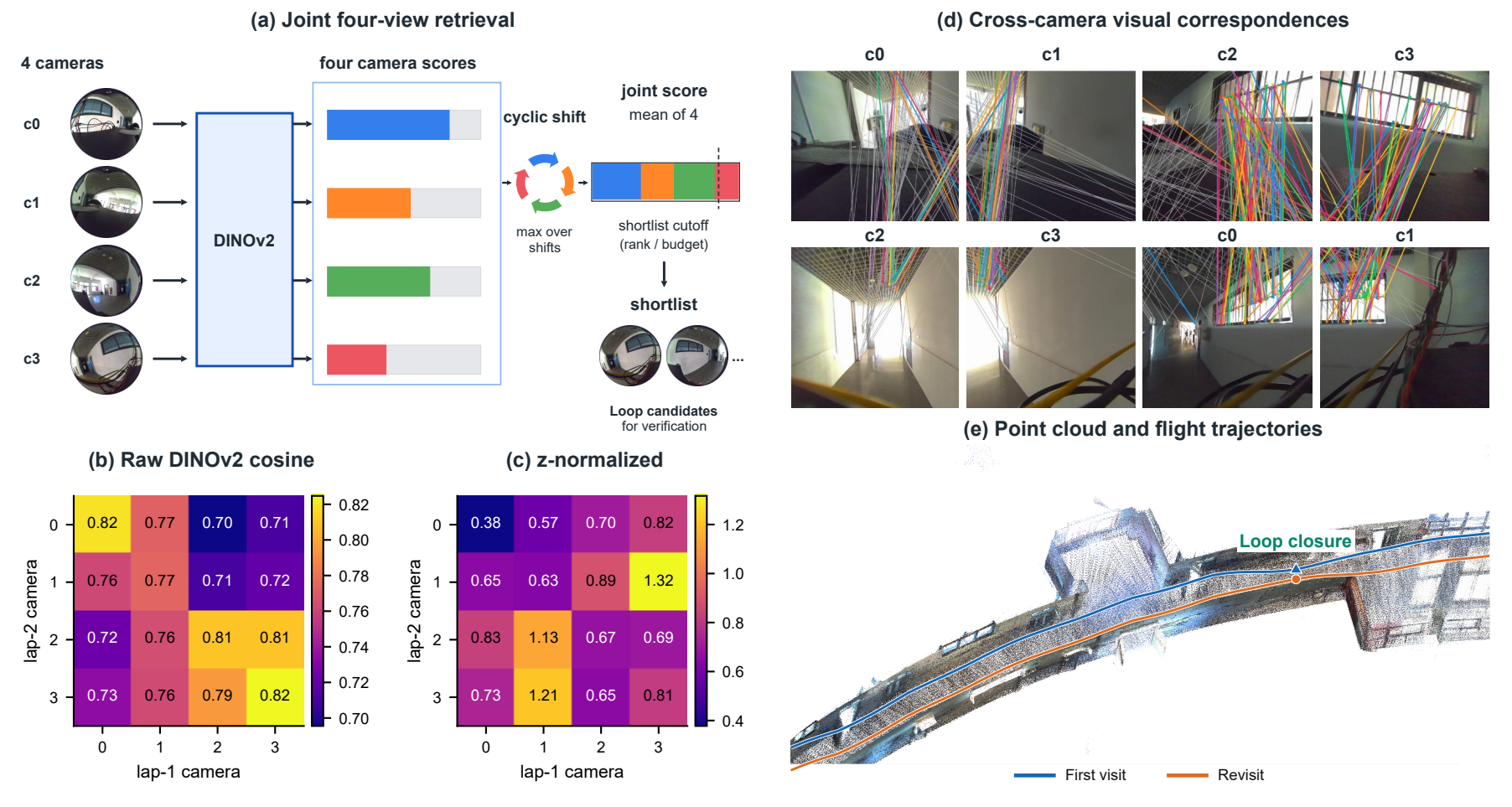

(a) Joint four-view retrieval
4 cameras
four camera scores
joint score
mean of 4
cyclic shift
c0
c1
c2
c3
DINOv2
max over shifts
shortlist cutoff (rank / budget)
shortlist
...
Loop candidates for verification
(d) Cross-camera visual correspondences
c0
c1
c2
c3
c2
c3
c0
c1
(b) Raw DINOv2 cosine
lap-2 camera
lap-1 camera
0.82 0.77 0.70 0.71
0.76 0.77 0.71 0.72
0.72 0.76 0.81 0.81
0.73 0.76 0.79 0.82
0.82 0.80 0.78 0.76 0.74 0.72 0.70
(c) z-normalized
lap-2 camera
lap-1 camera
0.38 0.57 0.70 0.82
0.65 0.63 0.89 1.32
0.83 1.13 0.67 0.69
0.73 1.21 0.65 0.81
1.2 1.0 0.8 0.6 0.4
(e) Point cloud and flight trajectories
Loop closure
First visit
Revisit